

SUMMARY TABLE: BIOLOGY GRADE 11

Sub-Strand 3.1: Reproduction in Animals — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 3.1: Reproduction in Animals
Driving Question	DRIVING QUESTION: How do different animals ensure successful fertilisation and development of offspring, and how can science assist when human reproduction fails?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Frogs, Hens and Cows: Many Ways to One Goal – How Do Animals Reproduce?	What did we observe about how frogs and fish release eggs and sperm in water, and how hens, snakes, cows and other animals were grouped as oviparous, viviparous or ovoviviparous?	What did we learn about why external fertilisation needs water and produces many eggs, while internal fertilisation protects fewer offspring on land?	How does this help explain why frogs release thousands of eggs while a cow produces one calf, and what new question do we have about the structures mammals need for internal fertilisation and how IVF fits in?
Lesson 2 Built for Reproduction: Dissecting Mammalian Reproductive Structures	We dissected a small mammal and used models and charts to observe the testes, epididymis, sperm duct and urethra in males, and the ovaries, oviducts, uterus, cervix and vagina in females. We noted their shape, position and texture.	Each part of the mammalian reproductive system is adapted to its function: the testes and ovaries produce gametes, tubes transport them, the oviduct is where fertilisation occurs, and the muscular, well-supplied uterus supports the developing embryo. Internal development protects a few offspring, unlike the frog's many eggs.	A cow can give birth to one well-protected calf because its reproductive structures keep the gametes safe, bring them together inside the body and support the embryo for months, whereas frogs near Kisumu rely on many eggs because external fertilisation is risky. This also helps explain how science such as IVF can assist when part of the pathway fails, but we still need to find out how sperm and egg meet and interact at the cellular level.
Lesson 3 Built for the Journey: Structure of Sperm and Ovum	What did you observe about the structure of the sperm cell and ovum (shape, size, tail, acrosome, mitochondria, jelly coat, food reserves) and about the number of gametes in frogs, hens, cows and humans?	What did you learn about how each structure helps movement, protection and fusion during fertilisation, and about what happens in IVF compared with natural fertilisation?	How can you use the structure of sperm and ovum to explain why frogs near Lake Victoria need thousands of eggs while cows produce one calf, and what do you still wonder about how the body controls the monthly release of the ovum?

Lesson 4 Hormonal Conductors: How Pituitary and Ovarian Hormones Regulate the Menstrual Cycle and Egg Release	Hormone charts and graphs showed FSH rising early in the cycle, oestrogen rising before a sharp LH surge near Day 14, ovulation shortly after the LH peak, and progesterone rising in the second half of the cycle, then falling before menstruation. Imbalance graphs differed from the normal pattern.	The pituitary gland (FSH and LH) and the ovaries (oestrogen and progesterone) interact through negative and positive feedback to time follicle development, ovulation and preparation of the uterine lining. Hormonal imbalances can disrupt ovulation and the cycle, and accurate knowledge helps to dispel myths and manage menstrual health.	Humans and cows usually release one egg at a time because hormone feedback loops control when a single follicle matures and ovulates, whereas frogs rely on large numbers of eggs released into water. When natural hormone regulation fails, biotechnology such as IVF can use hormone treatment to assist, but the baby still develops in the uterus. This leads to the next question: how does the hormonal balance shift if fertilisation occurs?
Lesson 5 From Fertilisation to Birth: How Hormones and the Placenta Sustain Pregnancy	We observed animations of fertilisation, implantation, placenta formation and birth, and heard a nurse/midwife explain antenatal care and stages of labour. We saw that the embryo implants in the uterine wall, the placenta forms, and the umbilical cord links the baby to the placenta.	We learned that hormones (hCG, progesterone, oestrogen, oxytocin, prolactin) coordinate pregnancy, birth and lactation, and that the placenta exchanges oxygen, nutrients and wastes between mother and baby without mixing their blood.	One well-nourished offspring (cow, human) is possible because the placenta and hormones provide sustained internal support, unlike frogs whose many eggs in water are unprotected. Our new question is: What happens when natural fertilisation or pregnancy cannot occur?
Lesson 6 Helping Where Nature Needs a Hand: Debating IVF, Surrogacy and Sperm Banking	We read case studies of couples with blocked fallopian tubes, low sperm count, hormonal imbalance, a need to store sperm before cancer treatment, and an absent uterus, and we heard scientific and ethical arguments in a structured debate.	IVF, surrogacy and sperm banking are biotechnologies that address specific structural or hormonal causes of infertility. In IVF fertilisation happens in a dish, but the embryo is transferred to the uterus and develops there. These options raise ethical questions about cost, consent, access and culture.	Just as frogs, hens and cows have different strategies to ensure fertilisation and development, humans can use biotechnology to replace or bypass the step that fails (e.g., fertilisation when tubes are blocked), but the embryo still needs the uterus and hormones to develop, so the baby does not grow wholly in a test tube.
Lesson 7 Shared Decisions, Hormonal Health and Biotechnology: Our Final Explanation of How Animals Reproduce	What we observed: external fertilisation in frogs and fish (many eggs released into water), internal fertilisation in hens, cows and humans, the structures of the male and female reproductive systems in the specimen or drawings, the animation of fertilisation, pregnancy and birth, and arguments in the class debate and from the guest speaker about joint reproductive choices and biotechnology.	What we learned: the types of fertilisation and development (external, oviparous, viviparous, ovoviviparous); how structure relates to function in reproductive systems; how hormones control the cycle, pregnancy, birth and lactation, and what undersecretion and oversecretion can cause; how IVF, surrogacy and sperm banks assist when reproduction fails; and why shared decisions by both parents matter.	What we can now explain: why frogs near Kisumu lay thousands of eggs while a cow bears one calf (site of fertilisation, losses, parental care); why IVF babies do not grow wholly in a test tube (fertilisation occurs outside, but the embryo develops in the uterus); and how animals ensure successful fertilisation and development, and how science and shared, informed decisions support human reproductive health.

ATTRIBUTION AND LICENSING

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